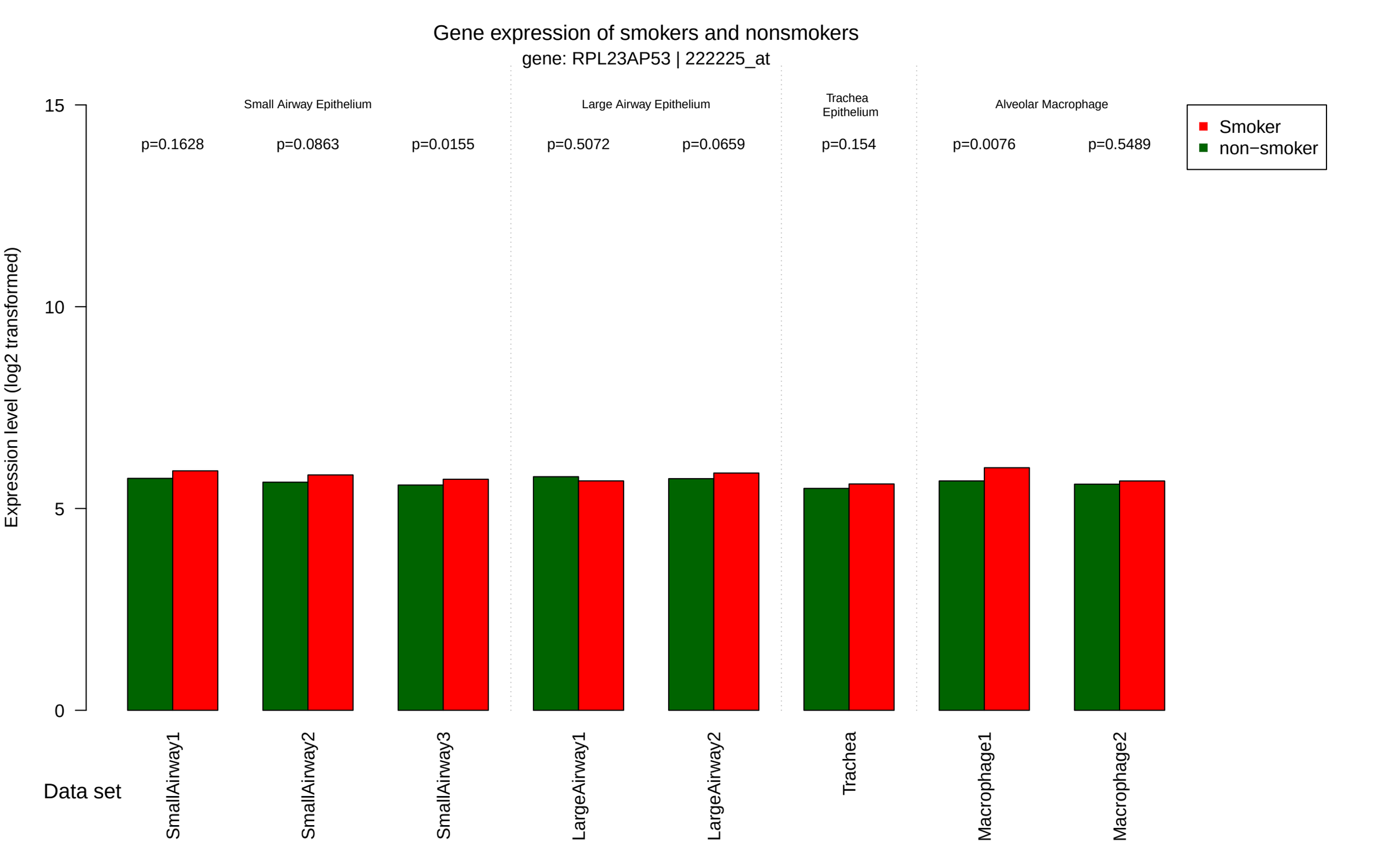
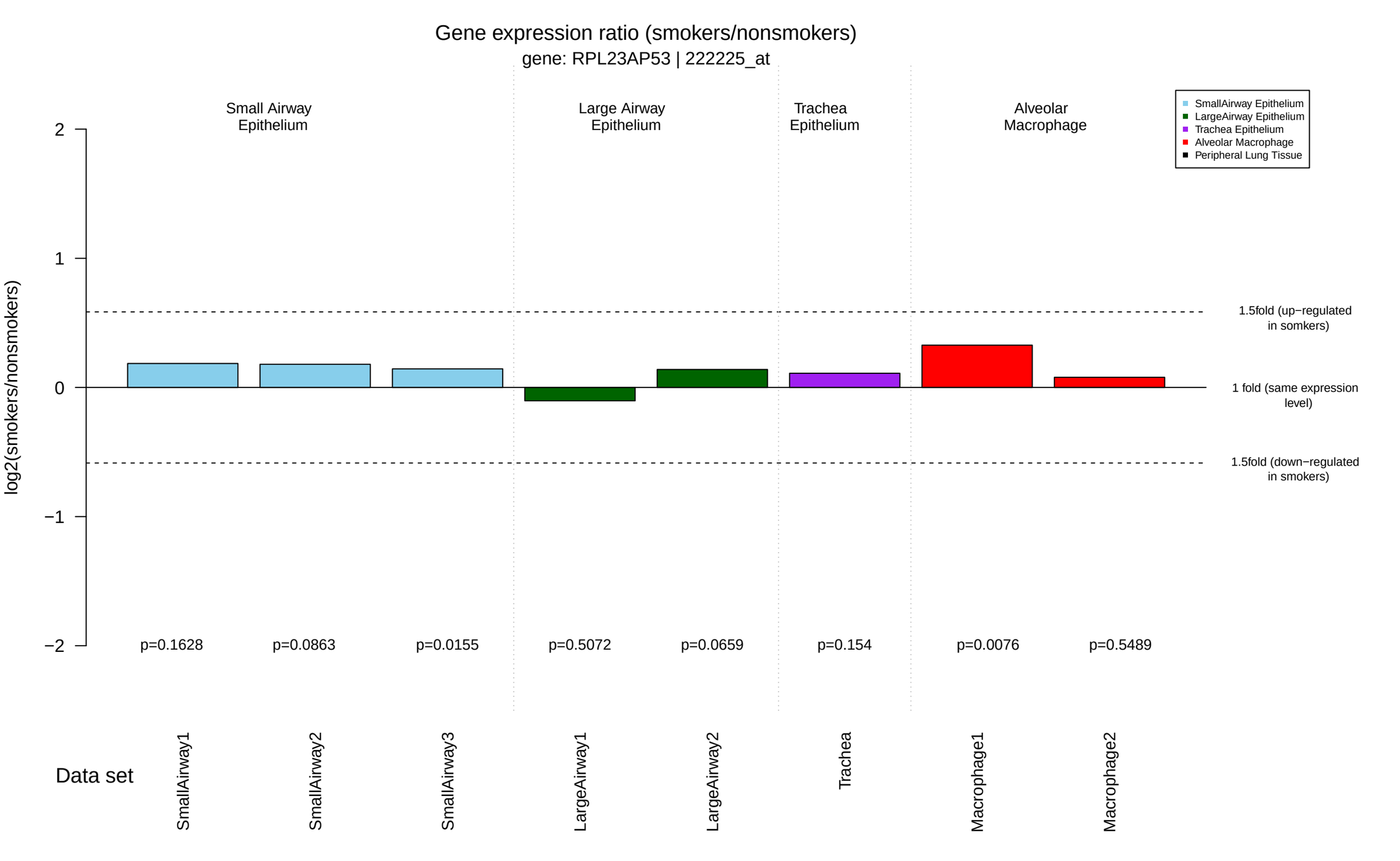
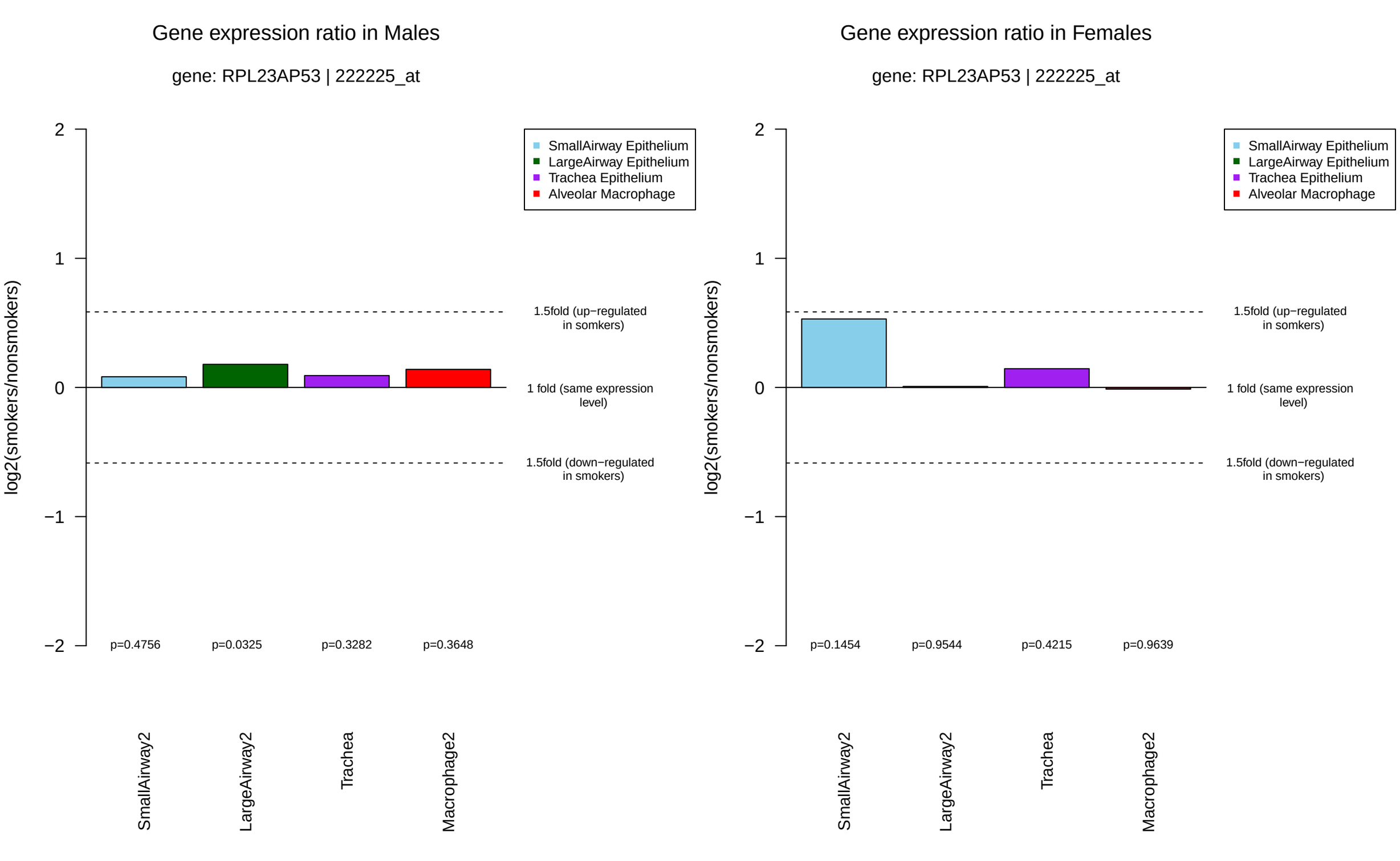
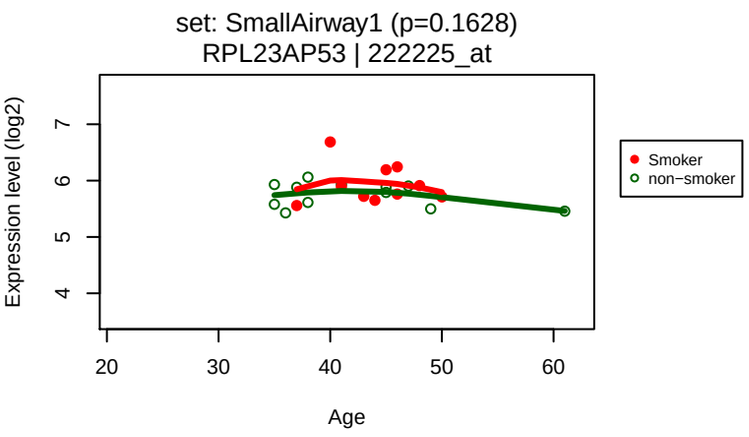
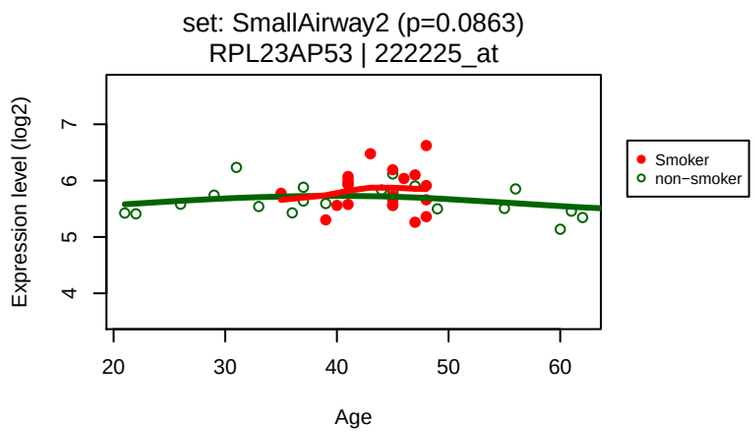
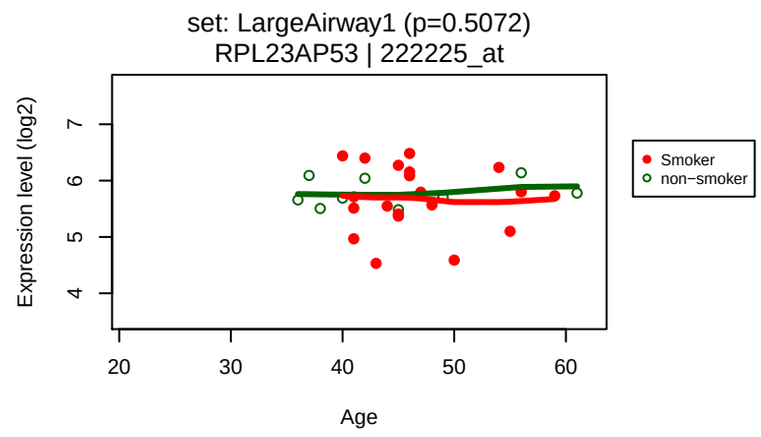

Gene expression of smokers and nonsmokers
gene: RPL23AP53 | 222225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1628
p=0.0863
p=0.0155
p=0.5072
p=0.0659
p=0.154
p=0.0076
p=0.5489
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

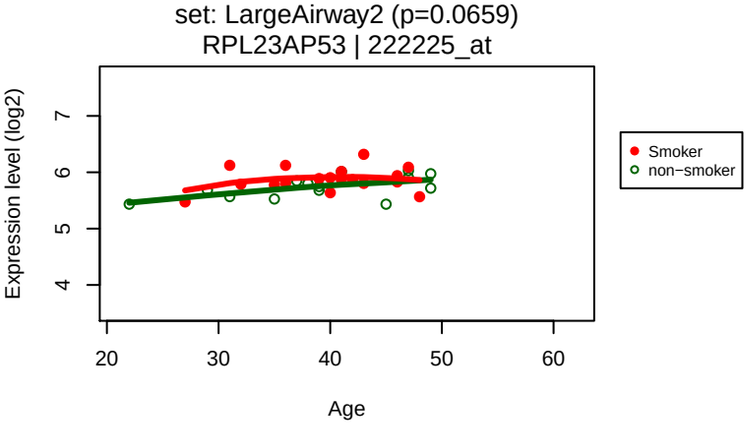

Gene expression ratio (smokers/nonsmokers)
gene: RPL23AP53 | 222225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1628
p=0.0863
p=0.0155
p=0.5072
p=0.0659
p=0.154
p=0.0076
p=0.5489
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

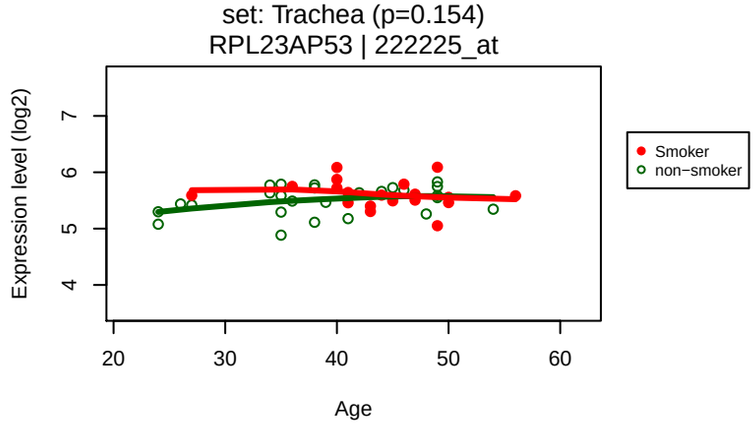

Gene expression ratio in Males
gene: RPL23AP53 | 222225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4756
p=0.0325
p=0.3282
p=0.3648
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL23AP53 | 222225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1454
p=0.9544
p=0.4215
p=0.9639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

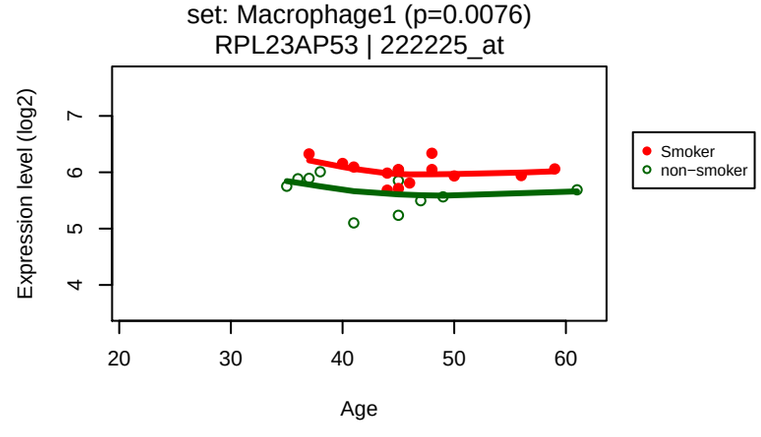

set: SmallAirway1 (p=0.1628)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker

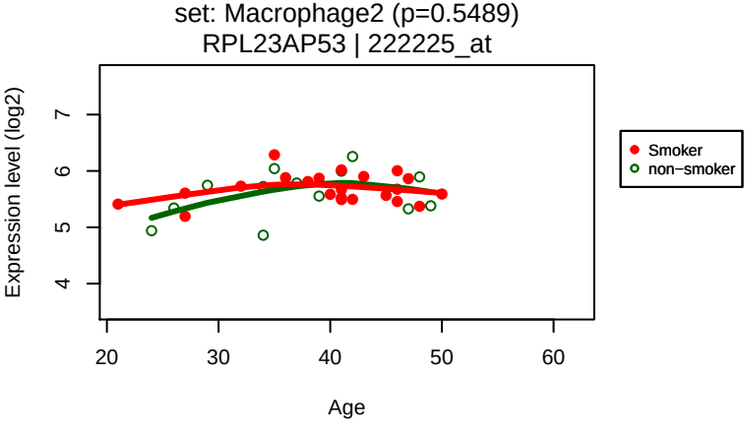

set: SmallAirway2 (p=0.0863)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker

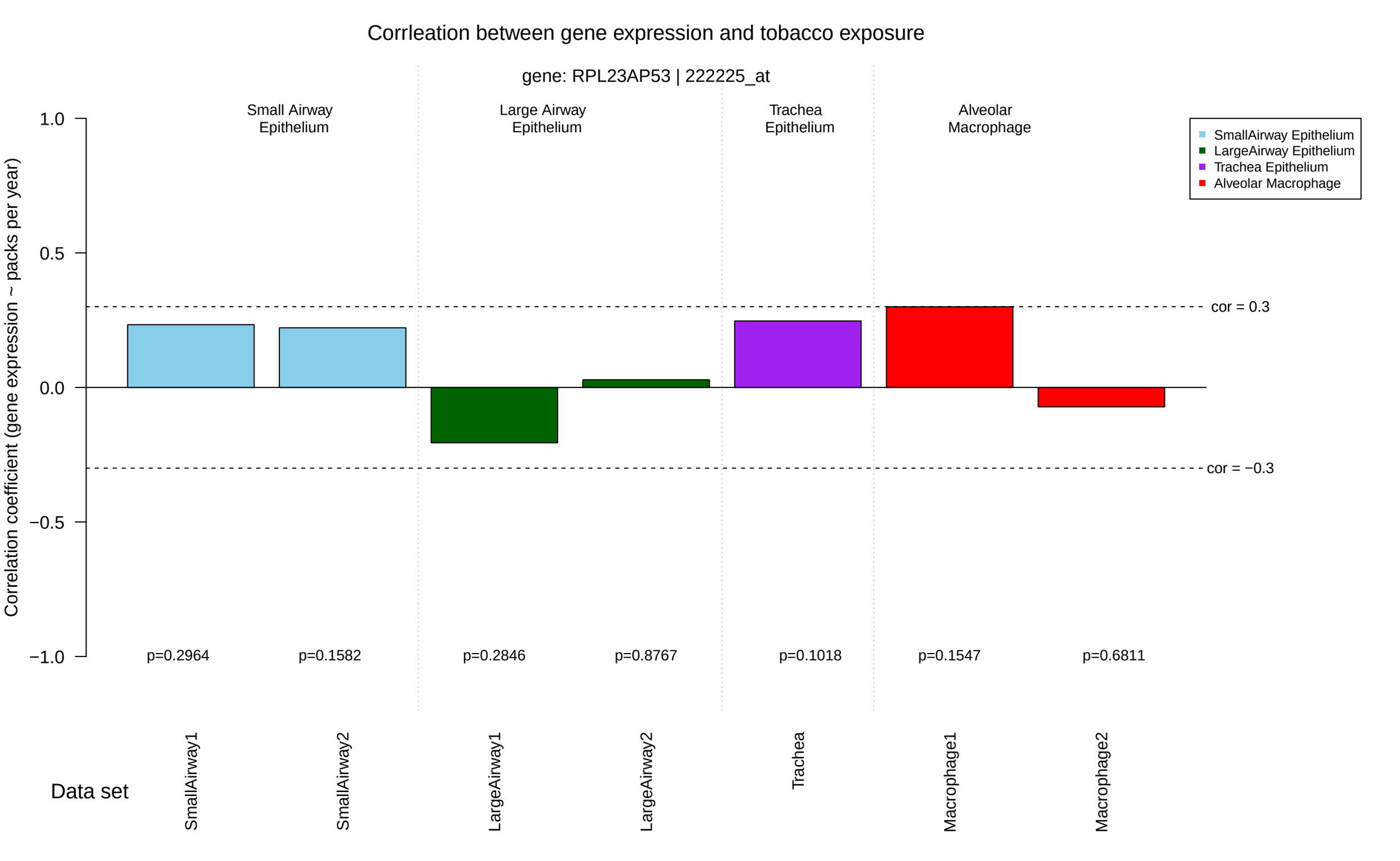

set: LargeAirway1 (p=0.5072)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0659)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.154)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0076)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5489)
RPL23AP53 | 222225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL23AP53 | 222225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2964
p=0.1582
p=0.2846
p=0.8767
p=0.1018
p=0.1547
p=0.6811
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2